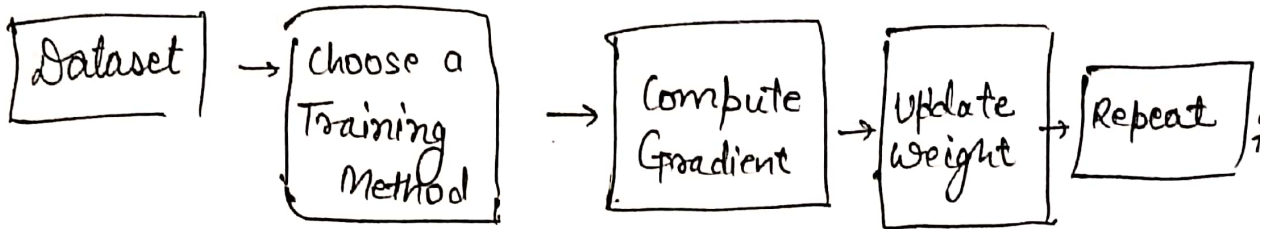


* Batch Vs Mini-Batch vs SGD.

all three are ways to compute Gradients and update model weight during training.



@codingdidi

① Batch Gradient Descent.

⇒ Use the Entire Dataset to compute Gradient once per update.

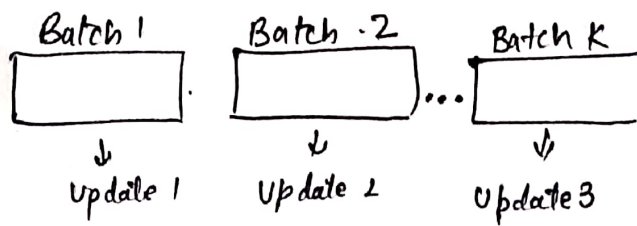


- How it works
- Take all N sample
 - Compute loss
 - Compute Gradient
 - Update weight once
 - Repeat

Dataset = 1000 Samples →
1 update per epoch.

② Mini-Batch

⇒ Use a small Batch of data to compute gradient and update. More common in practice.

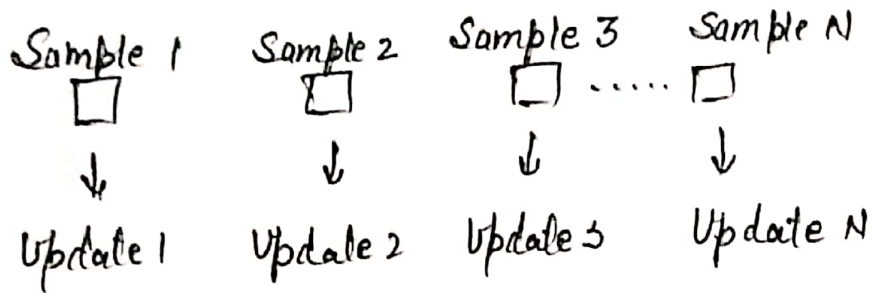


- How it works?
- * Split data into small batch of size B.
 - * For each batch:
 - Compute loss
 - Compute Gradient
 - update weights
 - * Repeat for all batches.

Dataset = 1000 Samples, Batch size (B) = 100 → $\frac{1000}{100} = 10$ updates per epoch.

③ Stochastic Gradient Descent (SGD)

⇒ Use One sample at a time to compute



How it works?

- * pick one sample
- * Compute loss
- * Compute Gradient
- * Update Weight.
- * Repeat

@rodbydd

In Deep Learning, Mini-Batch Gradient Descent is widely used!